

Module 1 – Institutional Requirements

UConn Standards Page: <https://kb.uconn.edu/space/IKB/10779852898/Accessibility>

Explain what UConn requires and how institutional policy connects to WCAG and federal requirements:

At UConn, there are certain requirements the entire university follows when it comes to developing or updating apps and websites. UConn's Digital Accessibility Policy requires its digital information, communication, content, and technology to be accessible to people with disabilities. This applies to the development, purchase, implementation, and maintenance of digital technology across the university.

These requirements are connected to a larger federal and state system of accessibility laws, federal disability-rights requirements such as the ADA and Section 504 require educational institutions to provide people with disabilities equal access to their programs, services, and activities, which UConn's policies reflect and establishes accessibility as a responsibility across the university, whether its administration, student.

WCAG provides technical guidelines that can be used to determine whether digital content is accessible. WCAG is not itself a federal law; instead, it provides specific, testable accessibility criteria. WCAG also has different levels of conformance—A, AA, and AAA, these are ratings that show how accessible your website is to different kinds of users, so simply saying that something is “WCAG compliant” does not explain its rating or the requirements it meets.

When evaluating a website or app at UConn, it is important to consider both the institutional requirements and the accessibility standards that apply.

For example, if a department is purchasing a new website tool and discovers that some of its features cannot be operated with a keyboard, the department should not assume the tool is accessible simply because it looks accessible. The keyboard barrier needs to be addressed because accessibility applies to the technology UConn purchases and uses, not just technology it develops itself.

A common misunderstanding is that WCAG is a federal law that every website must follow. In reality, WCAG is a technical standard created by the W3C. Different laws, institutions, and policies can require or use WCAG as an accessibility standard, while the legal requirements themselves come from the applicable laws and regulations.

Module 2 - Visual Cues and Meaning

Explain how color, shape, position, boundaries, and other visual cues can communicate meaning, and how that meaning should remain perceivable:

Visual cues are an essential part of web design; they allow users to identify what parts of a website are interactive, what content may be important, and can influence how the user flow is structured in a website or app. Color, shape, position, size, boundaries, and other visual characteristics can all communicate information to a user. However, designers need to make sure that important information is not dependent on a single visual cue that some users may not be able to perceive.

Color communicates meaning in many ways, but it is also easy to get wrong when it comes to web development. Not everyone has the same color perception, so color should not be the only way information is communicated. For example, using only red to show an incorrect answer and green to show a correct answer could make it difficult for some users to understand the difference. Adding text, an icon, or another visual indicator gives users another way to understand the information. Color contrast is also important because text needs enough contrast from its background to remain readable. WCAG requires a minimum contrast ratio of 4.5:1 for normal text and 3:1 for large text, with some exceptions. (AA Requirements)

A website uses red text to show that a form field contains an error.

Problem: A user who cannot distinguish the red may not know that the field contains an error.

Better design: Use an error message or icon along with the color.

The color can reinforce the meaning, but it should not be the only way the error is communicated.

Shape and position can also communicate meaning. For example, the shape of a button or icon can help users recognize what it does, while the position of an element can show how it relates to other content. However, a user should not have to rely only on a shape or location to understand an instruction. If an instruction says to "select the button on the right," someone who cannot perceive the visual arrangement may not know which button is being referenced. Providing a label or another identifying characteristic can make the same information easier to understand.

The way a page is laid out can also affect how users understand relationships between content. A website may place an image next to a description to show that the two are related, but that relationship should remain understandable when the layout changes. WCAG's reflow requirements allow content to rearrange when a user zooms in or views a page on a smaller screen, as long as important information and functionality are not lost. Similarly, text should be able to be enlarged to 200% without causing content or functionality to become unavailable.

This means that when using visual cues, it is important to ask: What information am I communicating visually, and would a user still understand that information if they could not perceive that specific visual cue or if the layout changed?

If the answer is no, another way of communicating the information may be needed. Tools such as WAVE can help identify potential accessibility problems when testing an actual webpage, but testing tools should be used alongside an understanding of accessibility requirements rather than as a replacement for them.

A common misunderstanding is that accessibility means avoiding visual cues such as color, shape, or layout altogether. This is not the case. Visual cues are useful and can make an interface easier to understand; the important part is making sure that essential information is not communicated through only one visual characteristic when users may not be able to perceive it. Accessibility means providing another way to understand the same information when necessary.

Module 3 - Images and Purpose

Explain how image purpose and context determine whether an image needs alt text, a decorative treatment, or another alternative:

Images can communicate information, support the meaning of a page, or simply be used for decoration. Because of this, not every image should be treated the same way when it comes to accessibility. When deciding how an image should be handled, it is important to ask what purpose the image serves and what information a user would miss if they could not see it.

Informative → Communicates information → Provide meaningful alternative

Decorative → Adds no meaningful information → Can use alt

Functional → Performs an action or leads somewhere → Describe the function/destination

Complex → Contains detailed information → May need a longer description or equivalent information

Informative images should have alternative text that communicates the important information the image provides, rather than simply describing everything that can be seen. Decorative images that do not add meaningful information can use an empty alt attribute, so they can be skipped by assistive technology. More complex images, such as charts, graphs, diagrams, or maps, may need a short description along with a longer explanation of the important information they contain.

The same idea applies to video and other media: important visual information that is not communicated through audio may need another alternative, such as audio description or a transcript.

When deciding what an image needs, the main question should be: What is this image trying to communicate, and what would a user miss without it?

For example, a photo of a campus building used only as decoration could be decorative, while the same photo could need alt text if the page is identifying that specific building. A common misunderstanding is that every image needs a detailed description. Accessibility is about providing the information a user needs, which sometimes means a detailed description, sometimes a short one, and sometimes no description at all.

Module 1 Sources:

<https://policy.uconn.edu/2019/08/02/digital-accessibility-policy/>

<https://www.w3.org/WAI/WCAG22/Understanding/conformance>

<https://www.w3.org/WAI/WCAG22/quickref/>

<https://www.ed.gov/laws-and-policy/civil-rights-laws/disability-discrimination/disability-discrimination-key-issues/disability-discrimination-technology-accessibility>

<https://www.w3.org/TR/WCAG22/>

Module 2 Sources:

<https://www.w3.org/WAI/WCAG22/Understanding/use-of-color.html>

<https://www.w3.org/WAI/WCAG22/Understanding/contrast-minimum.html>

<https://www.w3.org/WAI/WCAG22/Understanding/reflow.html>

<https://wave.webaim.org/extension/>

Module 3 Sources:

<https://www.w3.org/WAI/tutorials/images/decorative/>

<https://www.w3.org/WAI/tutorials/images/complex/>

<https://www.w3.org/WAI/media/av/>

<https://www.w3.org/WAI/media/av/description/>

<https://www.w3.org/WAI/tutorials/images/informative/>

